



RAFFLES INSTITUTION

2025 YEAR 6 TERM 3 TIMED PRACTICE

CANDIDATE
NAME

CLASS

25

COMPUTING

9569

Theory

2 hours

Additional materials: Answer Papers

READ THESE INSTRUCTIONS FIRST

Answer papers will be provided with the question paper.
Write your name and class on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.

Approved calculators are allowed.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.
The total number of marks for this paper is 70.

FOR EXAMINER'S USE					TOTAL
Q1	Q2	Q3	Q4	Q5	
					70

This document consists of **5** printed pages and **1** blank page.

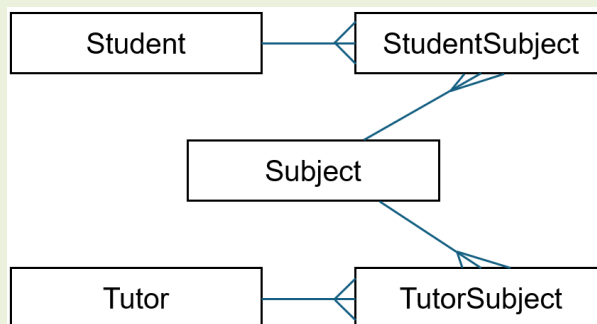
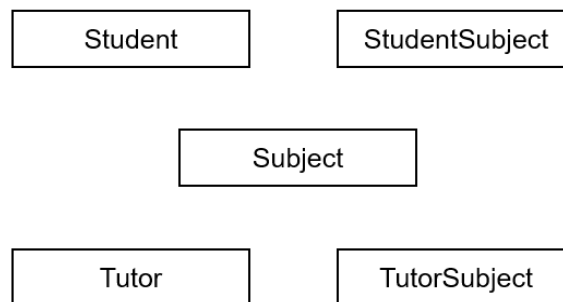
RAFFLES INSTITUTION
Mathematics Department

- 1 A junior college intends to create a centralised consultation booking system and has engaged a small team of H2 Computing students to develop it. The team decides to use a relational database to store relevant information.

First, the database needs to store information about students, including their names, class groups, email addresses and subjects they are taking, as well as information about subject tutors, including their names, departments, email addresses, and teaching information (subjects and the corresponding class groups).

Each subject tutor may be teaching more than one subject, and each subject is to be identified by its unique subject code (e.g. H2 Computing has subject code 9569).

- (i) Copy the following incomplete entity-relationship (E-R) diagram and complete it by adding four relationship lines. [2]



- (ii) Explain the need to include the `StudentSubject` and `TutorSubject` tables in the relational database. [2]

These are join tables, used to capture information describing many-to-many relationships. Here, for example, students can enroll in multiple subjects, and each subject can be taken by multiple students. Without a join table, there needs to be repeated data for every combination of student and subject, which leads to data redundancy. Therefore, join tables allow for multiple students to have multiple subjects without data redundancy.

Also, by using join table, it is possible for us to enforce referential integrity in this situation through the use of foreign keys, thus helping to maintain data integrity.

Next, the database should also store consultation booking information. To book a consultation on the system, the student would need to indicate the tutor, the subject and the consultation timing (e.g. 2024-10-15 15:30).

(iii) Two of the tables in the database have the following table descriptions:

Student (StudentID, StudentName, ClassGroup, StudentEmail)

Tutor (TutorID, TutorName, TutorDept, TutorEmail)

Write table descriptions for the other four tables, ensuring each table is in third normal form and indicating the primary keys and the foreign keys. For these tables, rely on composite keys where appropriate to uniquely identify each record. [7]

StudentSubject (StudentID, SubjectCode)

Primary Key: StudentID, SubjectCode

Foreign Keys: StudentID, SubjectCode

TutorSubject (TutorID, SubjectCode, ClassGroup)

Primary Key: TutorID, SubjectCode, ClassGroup

Foreign Keys: TutorID, SubjectCode

Subject (SubjectCode, SubjectName)

Primary Key: SubjectCode

Booking (StudentID, TutorID, SubjectCode, BookingTiming)

Primary Key: StudentID, BookingTiming

Foreign Keys: StudentID, TutorID, SubjectCode

(iv) State one possible database constraint that can be applied to the booking system to maintain data integrity. [1]

Any one of the following:

- A **UNIQUE** constraint to prevent duplicate bookings at the same time slot.
- A **FOREIGN KEY** constraint to ensure the StudentID exists in the Student table.
- A **NOT NULL** constraint to ensure BookingTiming must be provided.

(v) Write an SQL query to find the total number of bookings made per subject, and display the subject name along with the total number of bookings. Order the results by the number of bookings in descending order. [5]

```
SELECT
    subj.SubjectName,
    COUNT(*) AS TotalBookings
FROM
    Booking b
```

```

JOIN
  Subject subj ON b.SubjectCode = subj.SubjectCode
GROUP BY
  subj.SubjectName
ORDER BY
  TotalBookings DESC;

```

1m – SELECT
 1m – SUM
 1m – JOIN clause
 1m – GROUP BY
 1m – ORDER

(iv) Justify why it is better for the team to develop a web app instead of a native app for the college's consultation booking system. [2]

- Web app can be accessed from any device with a web browser, so any teacher or student can access the system using their mobile device or computer.
- Easier to develop and maintain a single web app, rather than having multiple native apps for different platforms. This is important as the development team of students are likely to have limited time and resources.

(v) Is a NoSQL database like MongoDB suited for the same application? Justify your answer. [2]

No, NoSQL is not recommended. The system requires structured data with strong relationships and constraints, which are better handled by relational (SQL) databases.

SQL supports data integrity and complex queries needed for booking management.

2 A media access control address (MAC address) is a unique identifier assigned to a network interface card for use as its network address within a network. A typical MAC address looks like this:

2C:54:91:88:C9:E3

A MAC address is represented by 12 hexadecimal numbers. For ease of reading, colons are used as a separator between every 2 numbers.

(a) Convert the hexadecimal number 2C:54:91:88:C9:E3 to a binary number, arranging the bits in groups of 4. Show your working clearly. [2]

Each hexadecimal digit is equivalent to 4 binary digits.

2: 0010 | C: 1100 | 5: 0101 | 4: 0100

9: 1001 | 1: 0001 | 8: 1000 | 8: 1000

C: 1100 | 9: 1001 | E: 1110 | 3: 0011

Hence, 2C:54:91:88:C9:E3₁₆ =

0010 1100 0101 0100 1001 0001 1000 1000 1100 1001 1110 0011₂

- (b) State the number of bits used to represent the MAC address. [1]

Total 48 binary bits

- (c) Describe how MAC addresses are used by networking devices during communication within a local area network (LAN). Give one specific example of a situation where MAC addressing is essential. [2]

- **[1 mark]:** Devices like switches use MAC addresses to forward data frames to the correct destination on a LAN.
- **[1 mark]:** For example, when a student prints a document, the switch uses the MAC address of the printer to deliver the data frame to the correct device.

- 3 A small office is developing a simple print management system to manage printers and their daily print jobs. A queue is implemented where the first print job to be sent to a printer will be the first job to be printed. They want to use Object-Oriented Programming (OOP) to model this situation.

For every printer the following data is recorded:

- a unique six-digit ID for the printer
- the model of the printer
- the location of the printer
- the list of print jobs
- the status of the printer (e.g.: "idle", "printing", or "offline").

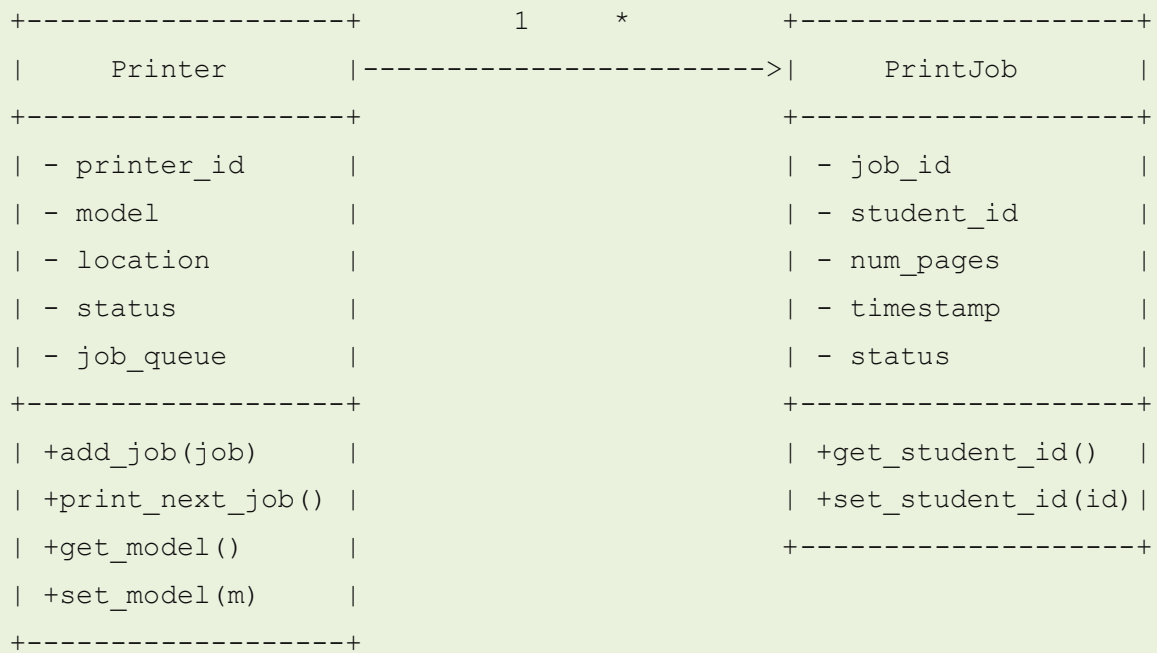
For every print job the following data is recorded:

- a unique three-digit ID for the print job
- the number of pages to be printed
- the timestamp when the job was submitted (i.e.: 25092024)
- the status of the print job (e.g.: "pending", "in progress", or "completed").

- (a) Draw two class diagram for the described situation, showing:

- any derived classes and relationship between the classes
- the properties needed in the classes
- suitable methods, in each class, to support the system.

Include one pair of get and set methods for one of the properties for every class. [8]



- (b)** Why is it important for the kiosk system to validate user input (e.g., student ID and number of pages) before accepting a print job? [1]

Validation is important in the printing kiosk system to ensure that the information entered by students, such as their student ID and the number of pages, is correct and follows expected rules. This helps prevent misuse, errors in billing, or failed print jobs due to invalid data.

- (c)** Identify and explain two suitable validation techniques that might be applied to the print job ID. [2]

1. Format Check (Student ID)

- Explanation: Ensures the student ID matches a specific format (e.g., 8-digit alphanumeric like S1234567).
- Why suitable: Prevents invalid or mistyped IDs that could be linked to the wrong student or cause print jobs to be rejected.

2. Range Check (Number of Pages)

- Explanation: Ensures that the number of pages requested is within an acceptable range (e.g., 1 to 50).
- Why suitable: Prevents accidental entries like "0" or "9999" pages, which could overload the printer or unfairly consume resources.

- 4 The current system uses a static data structure to implement the print queue. This limits the flexibility of the system because the queue has a fixed size.

(a) Describe how the system could be redesigned using a dynamic data structure. [4]

Linked list where the data of each node is a print job.

The head of linked list is the front of queue.

Enqueue adds node/ print job to the end of linked list.

Dequeue removes node/ print job from the head of the linked list.

(b) Explain **two** advantages of selecting a dynamic over static data structure. [4]

Any two points, with clear explanation.

1. No Fixed Size – Better Scalability

In a static data structure, the print queue has a fixed size defined at the start of the program. This can lead to situations where new print jobs are rejected if the queue is full, even if the system has unused memory.

A dynamic data structure, such as a linked list or dynamically resizable queue, can grow during runtime. This allows the system to accept any number of print jobs (limited only by available memory), ensuring that the queue remains functional even during peak usage periods like exam submission deadlines.

2. Efficient Insertion and Removal – Improved Performance

Static structures typically require shifting elements during insertion or deletion, especially in array-based queues. This operation becomes slower as the number of jobs increases.

Dynamic structures allow new print jobs to be added or removed by updating pointers, without shifting data. This leads to faster queue operations, which improves responsiveness in a self-service printing system. $O(1)$ operation
Optional to state the time complexities for this question.

3. Space Efficiency – Reduced Wasted Memory

Static structures allocate memory for the maximum possible queue size, even if only a few print jobs are active, leading to wasted memory.

Dynamic structures allocate memory only when needed, which results in better memory usage efficiency—especially useful in systems with multiple printers or limited resources.

(c) Explain a problem that might arise if a dynamic data structure is used in terms of memory allocation. [2]

Memory must be manually deallocated, by the computer system, when it is no longer needed.

Failure to do so leads to memory leak which causes out of memory error.

- 5 The university intranet can be accessed by all students, both locally and remotely, via the Internet. The network administrator has installed a firewall in the intranet.

(a) Describe two benefits that the intranet might provide for the students. [2]

- Can access from anywhere (geographical location) and anytime (time zone).
- Allow for collaboration among the users
- High security level within the campus; faster communication and collaboration for students and staff;
- Reduce the cost of connecting all devices to the Internet

(b) State the names of two kinds of malware that can infect a particular computer in the university network, and how the infection occurs. [4]

Any two of: Worm, virus, trojan, ransomware, spyware

Malware Type	What It Does	How Infection Occurs
Virus	Attaches to legitimate programs or files and spreads when they are opened. May damage or corrupt data.	Opening infected files from USB drives, email attachments, or shared folders.
Worm	Self-replicates and spreads automatically across networks without needing user action.	Exploits software vulnerabilities on connected systems (e.g., unpatched OS).
Trojan Horse	Appears as a legitimate application but provides a backdoor for attackers. Does not replicate.	Downloading and installing fake or pirated software.
Ransomware	Encrypts files and demands payment to restore access. Often causes data loss and operational disruption.	Clicking on phishing email links, downloading malicious attachments, fake updates.
Spyware	Silently monitors user activity and steals data such as keystrokes, passwords, or browsing history.	Bundled with free software or infects websites, installs without user awareness.

(c) State and explain two ways users can defend their computer against malware. [2]

Any two

1. **Install and regularly update antivirus or anti-malware software**
 - This helps detect, quarantine, and remove malware before it can cause harm.
2. **Avoid opening suspicious email attachments or downloading files from untrusted sources**
 - This reduces the risk of unknowingly activating malware such as Trojans or ransomware.
3. **Use antivirus software**
 - Antivirus software scans files and programs for known malware signatures and helps prevent, detect, and remove malicious software.
4. **Enable and configure a firewall**

- A firewall controls incoming and outgoing network traffic, blocking unauthorised access that could deliver malware to the system.

5. Keep operating systems and softwares up to date

- Security patches fix known vulnerabilities that malware could exploit.

(d) Describe one other non-malware threat and how it could compromise the university servers. Explain how the servers and the data stored on them could be protected and identify any limitation of the protection. [4]

Either one of the following or any other reasonable answer:

[1] description of a threat applied to the university

[1] the effect on the university

[1] describe how the university could protect against the identified threat

[1] identify a limitation of the protection

Phishing Attack

- A university staff member receives a fake email pretending to be from IT support, asking them to log in to a fraudulent portal.
- The attacker obtains login credentials and accesses sensitive student data or internal academic systems.
- Implement multi-factor authentication (MFA) and conduct regular cybersecurity awareness training for staff and students.
- MFA does not prevent users from being tricked into approving fraudulent login attempts, and training may not eliminate all human error.

Distributed Denial of Service (DDoS) Attack

- Attackers flood the university's online learning platform with excessive traffic, overwhelming its capacity.
- Students and staff are unable to access online lectures, course materials, or submit assignments during the disruption.
- Use DDoS mitigation services or firewalls with traffic filtering and rate-limiting rules.
- Mitigation tools can be costly and may occasionally block legitimate traffic, disrupting access for real users.

Any other reasonable non malware threats.

(e) Most of the students save their work in their laptops. Identify two possible dangers which may lead to the accidental loss of data in the local drives. Describe appropriate strategies to prevent such situations. [4]

Danger 1: Hardware Failure

Description: A laptop hard drive may become corrupted or fail due to age, overheating, or physical damage, leading to permanent data loss.

Prevention Strategy: Regularly back up files to external storage or cloud services (e.g., Google Drive, OneDrive) so that a copy is available even if the local drive fails.

Danger 2: Accidental Deletion or Formatting

Description: Students may accidentally delete important files or unintentionally format the drive.

Prevention Strategy: Enable file versioning or Recycle Bin features, and use data recovery software. Also, educate students to double-check before performing deletion or formatting actions.

- (d) Encryption and digital signature are widely used when students and staff send documents to each other. Explain the purpose of using these two in the network security. [2]

1.

Encryption ensures confidentiality by converting the document into unreadable text so that only authorised recipients with the correct decryption key can read it.

2.

Digital signature provides authentication and integrity, allowing the recipient to verify the sender's identity and confirm that the document has not been altered during transmission.

- (e) The existing network infrastructure is due for replacement and the University plans to provide cloud-based services to the students. Describe three benefits and two risks to the University if the cloud-based approach is adopted. [5]

Benefits

1. **Scalability:** Resources can be adjusted easily based on usage demand.
2. **Cost efficiency:** Reduced need for on-premise hardware and maintenance.
3. **Remote access:** Students can access services from anywhere, which supports hybrid learning.

Risks:

4. **Data privacy concerns:** Sensitive student and staff data may be stored on third-party servers.
5. **Service downtime:** Dependency on cloud service providers means any outage could disrupt access to critical services.